

Challenge (Habanero)

Q1. Simplify the expression: $\sqrt{16} + \sqrt{9}$

- (A) 7
- (B) 5
- (C) 4
- (D) 3
- (E) 1

Q2. What is the value of $\sqrt{64} - \sqrt{25}$?

- (A) 1
- (B) 3
- (C) 5
- (D) 7
- (E) 9

Q3. A gaming server uses $\sqrt{256}$ GB of RAM. How many GB does it use?

- (A) 12
- (B) 14
- (C) 16
- (D) 18
- (E) 20

Q4. A gamer's score is $\sqrt{81} + \sqrt{16}$ points. What is their score?

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q5. Simplify the expression: $\sqrt{49} \cdot \sqrt{9}$

- (A) 21
- (B) 42
- (C) 63
- (D) 7
- (E) 14

Q6. A coding project requires $\sqrt{100} - \sqrt{25}$ lines of code. How many lines does it require?

- (A) 5
- (B) 7
- (C) 9
- (D) 11
- (E) 15

Q7. Simplify the expression: $\frac{\sqrt{36}}{\sqrt{4}}$

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) 15

Q8. A game developer's data usage is $\sqrt{121} + \sqrt{16}$ MB. How many MB does the developer use?

- (A) $11 + 4$
- (B) $15 + 2$
- (C) $12 + 3$
- (D) $13 + 1$
- (E) $14 + 5$

Open Ended Questions (FRQ)

Q9. Simplify the expression: $\sqrt{144} \cdot \sqrt{25} - \sqrt{100}$

Q10. A gaming team's score is $\sqrt{400} - \sqrt{81}$. If they score $\sqrt{49}$ points in the next round, what will be their total score?

Chef's Tips

- Check for radical simplification
- Verify operations follow order of operations
- Encourage critical thinking with word problems